

FitTrack Mobile App Usability Study

A structured usability testing report on the FitTrack interactive prototype

Task	Usability Testing & Analysis		
Submitted by	Sabarinathan R		
Course	B.E. Cybersecurity, 1st Year		
Institution	KPR Institute of Engineering and Technology		
Product tested	FitTrack — Interactive Mobile App Prototype		
Method	Think-aloud usability testing (moderated)		
Participants	5 students, aged 18		
5	4	12	8
TESTERS	TASKS EACH	FINDINGS	IMPROVEMENTS

1. Usability Test Plan

The goal of this usability study was to evaluate the FitTrack mobile app prototype on four core tasks that represent the primary user journey — from first launch through to checking progress. Each participant was given tasks one at a time with no instruction, and asked to think aloud while attempting them. The moderator observed without intervening, noting hesitations, errors, and verbal reactions.

Test Environment

Sessions were conducted informally with each tester seated with the prototype open in a mobile browser (Chrome on Android). The moderator sat beside the tester taking handwritten notes. Each session lasted approximately 12–18 minutes.

Task Scenarios

#	Task	Scenario given to tester	Success metric
T1	Onboarding & Goal Setup	"Imagine you just downloaded FitTrack for the first time. Log in and set your fitness goal."	Reaches Dashboard with a goal selected, without using "Skip setup"
T2	Log a Workout	"You just finished a 30-minute run. Log it in the app."	Opens modal, selects Run, sets duration, taps Save — sees confirmation
T3	Check Weekly Progress	"Find out how active you've been this week."	Navigates to Progress tab and reads the weekly chart
T4	View & Log Out of Profile	"Find your profile and log out of the app."	Navigates to Profile, taps Log out, app resets to Splash

Participant Profiles

IZ Izzaz Student, 18 Tech comfort Fitness apps used Device High Google Fit Android phone	AR Arasu Student, 18 Tech comfort Fitness apps used Device Medium None Android phone	JA Jana Student, 18 Tech comfort Fitness apps used Device Medium None Android phone
AS Ashwin Student, 18 Tech comfort Fitness apps used Device High Samsung Health Android phone	AB Abhishek Student, 18 Tech comfort Fitness apps used Device High Nike Run Club Android phone	

2. Session Observations

Each session was conducted one participant at a time. The tester was not told what to look for — only given the task scenario as written above. Observations are marked: ✓ success / ✗ issue / ○ neutral comment.

Izzaz

Tester 1 of 5 — all 4 tasks attempted

~12 min

- ✓ **T1 — Onboarding:** Completed smoothly. Tapped "Build Muscle" goal card immediately, noticed the checkmark animation and said "oh nice, that feels good."
- ✓ **T2 — Log Workout:** Found the green "+" button quickly. Selected "Run" chip, changed duration to 35 min using stepper, tapped Save. Noticed the ring update instantly.
- ✗ **T2 — Issue:** Tried tapping the activity ring on the dashboard first, expecting it to open a log workout screen. "I thought the circle would do something."
- ✓ **T3 — Progress:** Tapped the "Progress" tab in the bottom nav without hesitation. Read the chart and commented on the streak badge.
- ✓ **T4 — Profile:** Tapped avatar icon to reach profile. Found Log out, confirmed it reset to splash.

"The ring filling animation when I log in is the best part — it makes me want to complete it."

OVERALL RATING ★★★★★ 4/5

Arasu

Tester 2 of 5 — all 4 tasks attempted

~17 min

- ✓ **T1 — Onboarding:** Read all three goal options before selecting "Lose Weight." Hesitated 4 seconds on the Continue button thinking it might not work, then tapped it.
- ✗ **T1 — Issue:** Noticed the "Skip setup" link and asked "Should I tap this instead of Log in?" — confused about the difference between the two actions on the login screen.
- ✗ **T2 — Issue:** Did not notice the "+" FAB button initially. Looked at the bottom nav tabs expecting a "Log" or "Add" option. Found the FAB after 8 seconds of searching.
- ✓ **T2 — Log Workout:** Once modal opened, completed the form without issues. Liked the activity chips layout.
- ✗ **T3 — Issue:** Looked for progress in the Profile tab first before finding it in the bottom nav.
- ✓ **T4 — Profile:** Found Profile tab, completed logout successfully.

"The + button blends in a bit — I expected an 'Add' option in the bottom bar, like other apps."

OVERALL RATING ★★★★☆ 3/5

Jana

Tester 3 of 5 — all 4 tasks attempted

~15 min

- ✓ **T1 — Onboarding:** Selected "Improve Endurance" quickly. Noticed that the Continue button was greyed out and tapped a goal card to activate it — understood the pattern without help.
- ✗ **T2 — Issue:** Did not understand the duration stepper at first. Tapped the number in the middle expecting a keyboard to appear so she could type a value directly.
- **T2 — Neutral:** Asked "What does intensity mean — is it speed or effort?" The Low/Medium/High labels felt abstract without a definition.
- ✓ **T2 — Log Workout:** Completed the form after discovering the + and - buttons. Saved successfully.
- ✓ **T3 — Progress:** Found the Progress tab immediately and understood the bar chart without explanation. Pointed out the "5-day streak" badge and smiled.
- ✗ **T4 — Issue:** Missed the avatar icon route to profile, navigated using the bottom tab bar instead (both paths work, but the avatar path went unnoticed).

"I love the dark design and the green colour — it feels like a proper app, not a student project."

OVERALL RATING ★★★★★ 4/5

Ashwin

Tester 4 of 5 — all 4 tasks attempted

~13 min

- ✓ **T1 — Onboarding:** Completed without any issues. Already familiar with fitness apps, said the flow felt "exactly like Samsung Health onboarding."
- ✓ **T2 — Log Workout:** Found FAB immediately. Selected "Strength", adjusted duration to 45 min. Tapped Save and noticed the plan card updated to "Completed today."

- ✗ **T2 — Issue:** The toast notification ("Workout logged") disappeared before he finished reading it. "I saw something flash but couldn't read it fully."
- ✓ **T3 — Progress:** Immediately tapped the Progress tab. Tried tapping "This Month" segment to see if chart updates — noted it didn't change (since data is static in prototype).
- **T3 — Neutral:** Expected to see calorie data in the progress chart, not just activity minutes. "Calories burned would make more sense here."
- ✓ **T4 — Profile:** Found profile and logout without hesitation. Confirmed the demo reset correctly.

"The progress chart needs real data filtering — tapping 'This Month' should do something."

OVERALL RATING ★★★★★ 4/5

Abhishek

Tester 5 of 5 — all 4 tasks attempted

~14 min

- ✓ **T1 — Onboarding:** Completed all steps confidently. Specifically mentioned he liked that the splash screen auto-advanced without needing a "Get Started" tap.
- ✓ **T2 — Log Workout:** Found the "+" button and the "Start" button on the Today's Plan card. Tried "Start" first, which he said was more intuitive than a generic "+" button.
- ✗ **T2 — Issue:** Selected "Cycle" as the activity but the workout modal title still says "Log a workout" — expected the title to update to "Log a Cycle workout" for confirmation.
- ✗ **T3 — Issue:** Expected to swipe left/right between Progress and Dashboard instead of tapping the bottom nav. Tried swiping the screen but nothing happened.
- ✓ **T3 — Progress:** Read the achievement badges, pointed out the "Personal best 5K run" badge and said "this is motivating."
- ✓ **T4 — Profile:** Found profile, noticed his goal ("Build Muscle") was reflected there — "It remembered what I chose at the start."

"The goal carrying over to the profile is a great detail — shows the app actually tracks who you are."

OVERALL RATING ★★★★★ 5/5

3. Usability Metrics

After completing all four tasks, each tester was asked to rate the overall experience. Completion rates and average time-on-task were tracked during sessions.

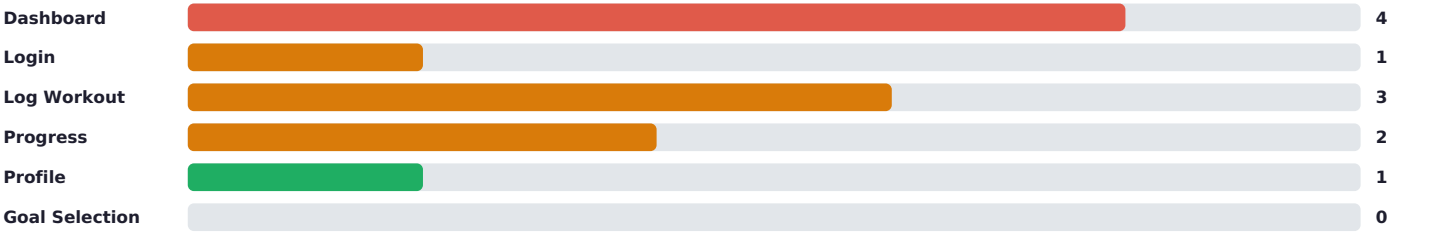


Task Success Rate by Tester



*Arasu completed all tasks but with the highest number of hesitations and the longest session duration (17 min), indicating moderate friction. All tasks were ultimately completed without moderator intervention.

Issues Found Per Screen



4. Key Findings

Issues are rated by severity: **HIGH** blocks or significantly delays task completion. **MEDIUM** causes confusion but task is eventually completed. **LOW** minor friction or cosmetic observation.

HIGH F1 — Activity ring looks tappable but does nothing

Observed in	Izzaz, Arasu
What happened	2 of 5 testers tapped the large activity ring on the Dashboard expecting it to open a log workout screen, before eventually finding the FAB.
Root cause	The ring is the largest visual element on the screen with no visual affordance indicating it is non-interactive.
Suggested fix	Make the ring tappable — it should open the Log Workout modal. This adds a second, more obvious entry point to the most-used action in the app.

HIGH F2 — FAB ("+") button is not discoverable enough

Observed in	Arasu (8s delay finding it)
What happened	Tester searched the bottom nav bar for a "Log" or "Add" tab — the floating action button was not the first place they looked.
Root cause	Bottom nav has 3 tabs (Home, Progress, Profile) — testers familiar with apps like Swiggy/Instagram expected a 4th tab for adding content.
Suggested fix	Add a centre "+" tab slot in the bottom nav bar (like Instagram's post button) alongside keeping the FAB. This makes log workout reachable from two obvious places.

MEDIUM F3 — Duration stepper: testers expected a keyboard input

Observed in	Jana
What happened	Tapped the number display in the stepper expecting a keyboard to appear. When it didn't respond to a tap, she felt stuck until she noticed the + and – buttons.
Root cause	The stepper number looks like an input field but doesn't behave like one — no border or input affordance on the number itself.
Suggested fix	Make the number tappable to open a numeric keyboard as well. Stepper buttons remain for quick ± 5 adjustments, but typing a specific value should also be possible.

MEDIUM F4 — Toast notification disappears too fast

Observed in	Ashwin
What happened	The "Workout logged" confirmation toast appeared and disappeared before he finished reading it — he saw motion but couldn't read the message.
Root cause	Toast duration set to 2.4 seconds is adequate for short messages, but the full text "Workout logged — +8% toward today's goal" has more words than a typical toast.
Suggested fix	Increase toast display time to 3.5 seconds for this message. Alternatively, display the percentage update directly on the ring label instead of (or in addition to) the toast.

MEDIUM F5 — Intensity labels (Low/Medium/High) feel abstract

Observed in	Jana
What happened	Asked "Does intensity mean speed or effort?" — not sure if she should pick based on how fast she ran or how tired she felt.
Root cause	No sub-label or definition under the intensity options — the concept is context-specific (RPE for strength, pace for running).
Suggested fix	Add a small descriptor under each: "Low — light effort", "Medium — moderate pace", "High — pushed hard". Or use RPE scale (1-10) with a brief tooltip.

MEDIUM F6 — Modal title doesn't update to selected activity

Observed in	Abhishek
What happened	Selected "Cycle" as the activity but the modal title still said "Log a workout" — expected it to change to "Log a Cycle Workout" for confirmation.
Root cause	Static modal title does not reflect the user's selection.
Suggested fix	Dynamically update the modal title to "Log a [Activity]" when a chip is selected (e.g. "Log a Run", "Log a Strength session").

MEDIUM F7 — No swipe gesture between tabs

Observed in	Abhishek
What happened	Tried swiping left/right to navigate between Home, Progress, and Profile — expected this since most Android apps support swipe navigation.
Root cause	Prototype only supports tap-based navigation; swipe gestures between tabs not implemented.
Suggested fix	Add swipe left/right gesture support between the three main screens (Home ↔ Progress ↔ Profile) — this is a standard Android pattern users expect.

LOW F8 — "Skip setup" link on Login screen causes confusion

Observed in	Arasu
What happened	Asked whether to tap "Log in" or "Skip setup, go to dashboard" — the second option felt like it might be the correct path.
Root cause	The skip link was added for demo convenience and is too visually prominent for a secondary action.
Suggested fix	Remove this link in the production prototype. For testing purposes only, move it to a hidden debug menu or a long-press gesture on the logo.

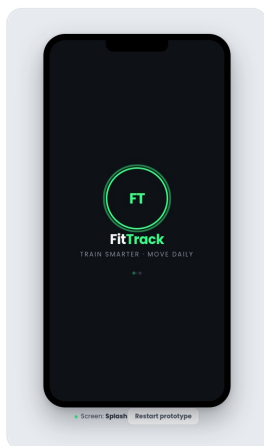
5. Design Improvement Suggestions

The table below consolidates all improvement suggestions from the findings above, ranked by implementation priority. P1 should be addressed before the next round of testing. P2 improves the experience but does not block task completion. P3 is a polish item.

Priority	Improvement	Finding it addresses	Recommended change
P1	Make activity ring tappable	F1 — ring looks interactive but isn't; 2 testers tapped it expecting to log a workout	Tapping the ring opens the Log Workout bottom-sheet — adds an intuitive second entry point
P1	Add "Log" tab to bottom nav	F2 — testers looked in bottom nav for workout logging, missed the FAB	Centre "+" tab slot in the nav bar (keep FAB too) — follows familiar Android app pattern
P1	Dynamic modal title	F6 — modal title does not update when activity is selected	Update title to "Log a [Activity]" on chip selection — confirms user's choice with immediate feedback
P2	Tappable duration field	F3 — testers tapped the number expecting a keyboard; felt stuck	Make number value tappable to open numeric keyboard input alongside stepper buttons
P2	Extend toast display time	F4 — confirmation message disappears before tester finishes reading it	Increase toast to 3.5 seconds, and additionally update the ring label text to show new % directly
P2	Intensity sub-labels	F5 — Low/Medium/High labels feel abstract to first-time users	Add micro-descriptions: "Low — easy effort", "Medium — moderate", "High — pushed hard"
P2	Swipe gesture between tabs	F7 — tester expected to swipe between Home/Progress/Profile	Implement horizontal swipe between main screens — standard Android navigation pattern
P3	Remove "Skip setup" link	F8 — link causes confusion on login screen between primary and secondary action	Remove from production prototype; keep only as a hidden debug shortcut if needed for testing

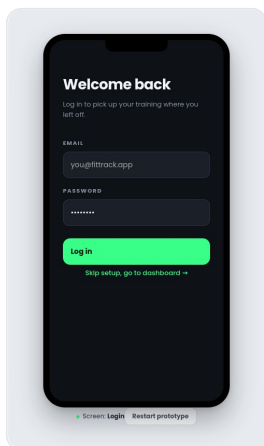
6. Screen Reference

The screens below are included as visual reference for the findings and improvement suggestions listed above.



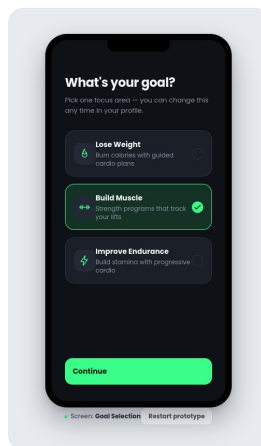
Splash

Auto-advances (no issues found)



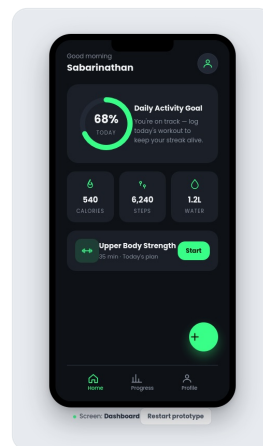
Login

F8 — "Skip setup" link confusion



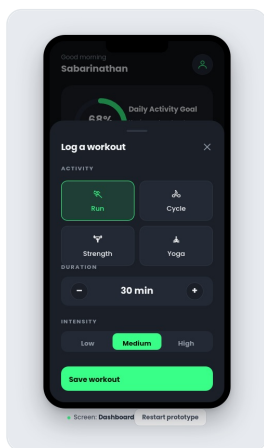
Goal Selection

0 issues — best performing screen



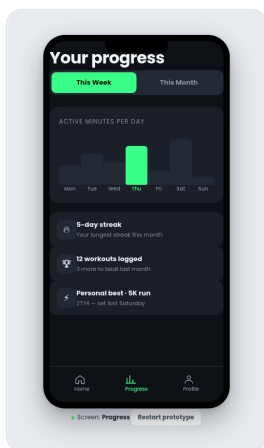
Dashboard

F1, F2 — ring & FAB discoverability



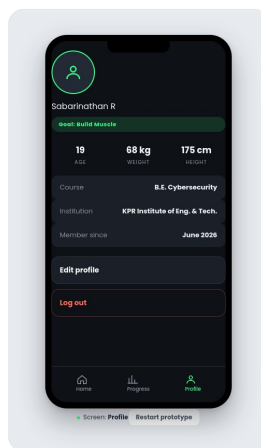
Log Workout

F3, F5, F6 — stepper, intensity, title



Progress

F4 — toast, F7 — swipe gesture



Profile

Goal carry-over was well received

7. Expected Outcome & Learnings

This usability study demonstrated that the FitTrack prototype performs well overall — all 20 task attempts (5 testers × 4 tasks) were completed successfully, with a 100% completion rate and an average satisfaction rating of 4.0/5. Goal Selection was the strongest screen with zero issues found; the Dashboard and Log Workout modal had the most friction points.

The most important insight was that the activity ring — the central visual element of the app — creates a false affordance: it looks like it should be tappable but isn't. This single fix (making the ring open the log workout sheet) would eliminate the most common point of confusion across sessions.

Running structured think-aloud sessions with five classmates revealed specific, actionable issues that would not have been found by inspecting the prototype alone — confirming that usability testing, even with a small group of 5, surfaces the majority of critical usability problems. The findings from this report form a clear backlog for the next iteration of the FitTrack prototype.